Teaching notes and solutions

Teacher notes

Test 5 and the second mock exam come before S73: do not ask for Bienaymé–Chebyshev or the law of large numbers in them. For the variance, have students circle the independence assumption. For concentration, distinguish the exact probability from the bound, and a sufficient rank from the true minimal rank. A simulation is not a proof, and an observed mean does not necessarily get closer with each new trial.

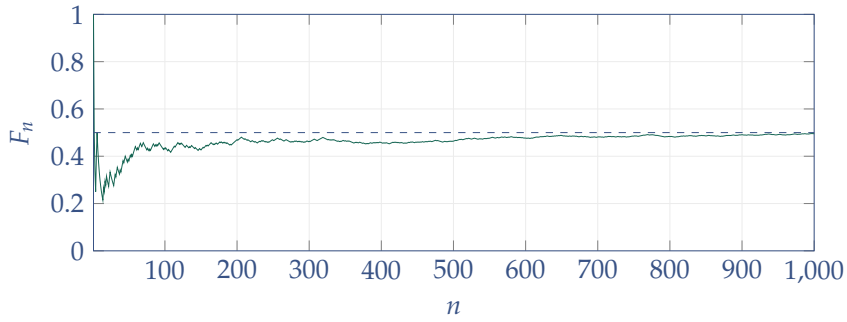
Solutions to the activity

1.

```
from random import Random

def trajectoire_piece(N=1000, graine=2026):
    rng = Random(graine)
    piles, frequences = 0, []
    for n in range(1, N+1):
        piles += rng.random() < 0.5
        frequences.append(piles/n)
    return frequences
```

Plot the points (n, F_n) with the list obtained. After two tosses giving one head and one tail, $F_2 = \frac{1}{2}$. Whatever the third result, F_3 is $\frac{1}{3}$ or $\frac{2}{3}$: the deviation from $\frac{1}{2}$ increases. Stabilisation is not a monotonic decrease of the deviation along each trajectory.



A simulated trajectory, seed 2026; the line is $y = \frac{1}{2}$.

2. Let B_i be the indicator of heads. Then $F_n = \frac{B_1+\cdots+B_n}{n}$. Each B_i has expected value $\frac{1}{2}$ and variance $\frac{1}{4}$. By linearity, $E(F_n) = \frac{1}{2}$; by independence and rescaling, $V(F_n) = \frac{n(\frac{1}{4})}{n^2} = \frac{1}{4n}$.

3. Bienaymé–Chebyshev gives $P(|F_{1000} - \frac{1}{2}| \geq 0.05) \leq \frac{1}{4\times 1000\times 0.05^2} = 0.1$. This is an upper bound, not the exact probability.

4. To ensure a complementary probability of at most 0.01, it suffices that $\frac{1}{4n\times 0.05^2} \leq 0.01$. This is equivalent to $n \geq 10000$. Then $P(|F_n - \frac{1}{2}| < 0.05) \geq 0.99$.

5. The simulation illustrates one finite realisation; other seeds give other trajectories. The inequality establishes a guarantee on the probability, for any sample satisfying the assumptions of independence and common distribution.

E1. The affine transformation gives $E(4X-1) = 4E(X) - 1 = 11$ and $V(4X-1) = 4^2V(X) = 32$. The standard deviation is $\sqrt{32} = 4\sqrt{2}$; the shift -1 does not change the dispersion.

E2. By linearity, $E(2X-Y) = 2E(X) - E(Y) = 4-5 = -1$. The independence of X, Y allows us to add the variances of $2X$ and $-Y$: $V(2X-Y) = 4V(X) + V(Y) = 4+4 = 8$.

E3. The mean of 64 independent variables with the same distribution keeps the expected value 10. Its variance is $\frac{4^2}{64} = \frac{1}{4}$, so its standard deviation is $\frac{1}{2}$.

E4. Bienaymé–Chebyshev with $\delta = 6$ gives $P(|X-100| \geq 6) \leq \frac{V(X)}{6^2} = \frac{9}{36} = \frac{1}{4}$. This bound gives neither an equality nor an exact value.

E5. We have $P(|M_n - \mu| \geq 0.2) \leq \frac{4}{n \times 0.2^2} = \frac{100}{n}$. For this bound to be at most 0.05, it suffices that $n \geq 2000$. The complement then has probability at least 0.95.

Solutions to Homework 14

1. Let B_i be the independent indicators of the successes. $F_n = (\frac{1}{n}) \sum_{i=1}^{n} B_i$, with $E(B_i) = p$, $V(B_i) = p(1-p)$. Hence

$$E(F_n) = \frac{1}{n}\sum_{i=1}^{n} p = p, \qquad V(F_n) = \frac{1}{n^2}\sum_{i=1}^{n} p(1-p) = \frac{p(1-p)}{n}.$$

2. $p(1-p) = p - p^2 = \frac{1}{4} - (p - \frac{1}{2})^2 \leq \frac{1}{4}$. Equality is attained for $p = \frac{1}{2}$.
3. For all $p \in [0,1]$, $P(|F_n - p| \geq 0.02) \leq \frac{1}{4n \times 0.02^2}$. It suffices that this bound be at most 0.05, that is $n \geq 12500$. By complement, this size guarantees the required probability, uniformly in p.
4. On $[0, 0.1]$, the function $p \mapsto p(1-p)$ is increasing because its derivative $1-2p$ is positive. Hence $p(1-p) \leq 0.1 \times 0.9 = 0.09$. It now suffices that $n \geq \frac{0.09}{0.05 \times 0.02^2} = 4500$. These sizes are sufficient in the sense of the bound, not necessarily minimal for the exact probability.
5. If all the trials copy the same variable B, then $F_n = \frac{B + \cdots + B}{n} = B$. Hence $V(F_n) = p(1-p)$, with no division by n. Adding the variances was allowed only because of independence; this assumption disappears here. For $0 < p < 1$, increasing the number of copies brings no new information.

Solutions to the textbook or past-paper work

M_N is the mean number of universal donors per sample of 100 people. Linearity gives $E(M_N) = 7.14$; independence gives $V(M_N) = \frac{6.63}{N}$.

$$P(|M_N - 7.14| \geq 0.14) \leq \frac{6.63}{0.14^2 N}.$$

To make this bound at most 0.05, it suffices to take

$$N \geq \left\lceil \frac{6.63}{0.05 \times 0.14^2} \right\rceil = 6766.$$

By complement, $P(7 < M_N < 7.28) \geq 0.95$.